

NLP Project Proposal

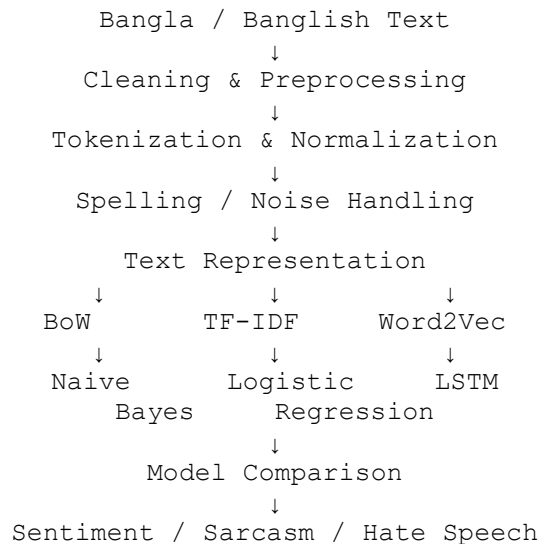
by
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Title: Context-Aware Bangla Text Analyzer for Sentiment, Sarcasm, and Hate Speech Detection

1. Background & Motivation

The increasing use of Bangla and Banglish on social media has created a need for automated systems to understand user-generated text. Such text often contains spelling variations, informal expressions, emojis, and context-dependent language. Detecting sentiment, sarcasm, and hate speech is therefore challenging using simple keyword-based approaches.

2. Proposed NLP Pipeline



The system will use Regex, tokenization, stop-word removal, normalization, and Levenshtein Edit Distance for preprocessing. BoW and TF-IDF will be used as baseline representations, while Word2Vec will be explored for semantic representation. Naive Bayes and Logistic Regression will provide traditional classification baselines, and an LSTM-based model will be investigated for capturing sequential and contextual information.

3. Input & Expected Output

Input:

“বাহ! কী অসাধারণ service, তিন ঘণ্টা অপেক্ষা করেও কাজ হলো না!”

Output:

Sentiment: Negative

Sarcasm: Yes

Hate Speech: No

4. Dataset & Evaluation

Publicly available Bangla datasets related to sentiment, sarcasm, and hate speech will be collected and prepared for experimentation. Where feasible, a small Bangla/Banglish evaluation corpus may also be prepared. The models will be compared using accuracy, precision, recall, F1-score, and confusion matrix.

5. Expected Outcome

The expected outcome is a functional Bangla/Banglish text analyzer that can identify sentiment, sarcasm, and hate speech from user input. The project will also provide a comparative analysis of traditional, embedding-based, and sequence-based NLP approaches for Bangla text analysis.